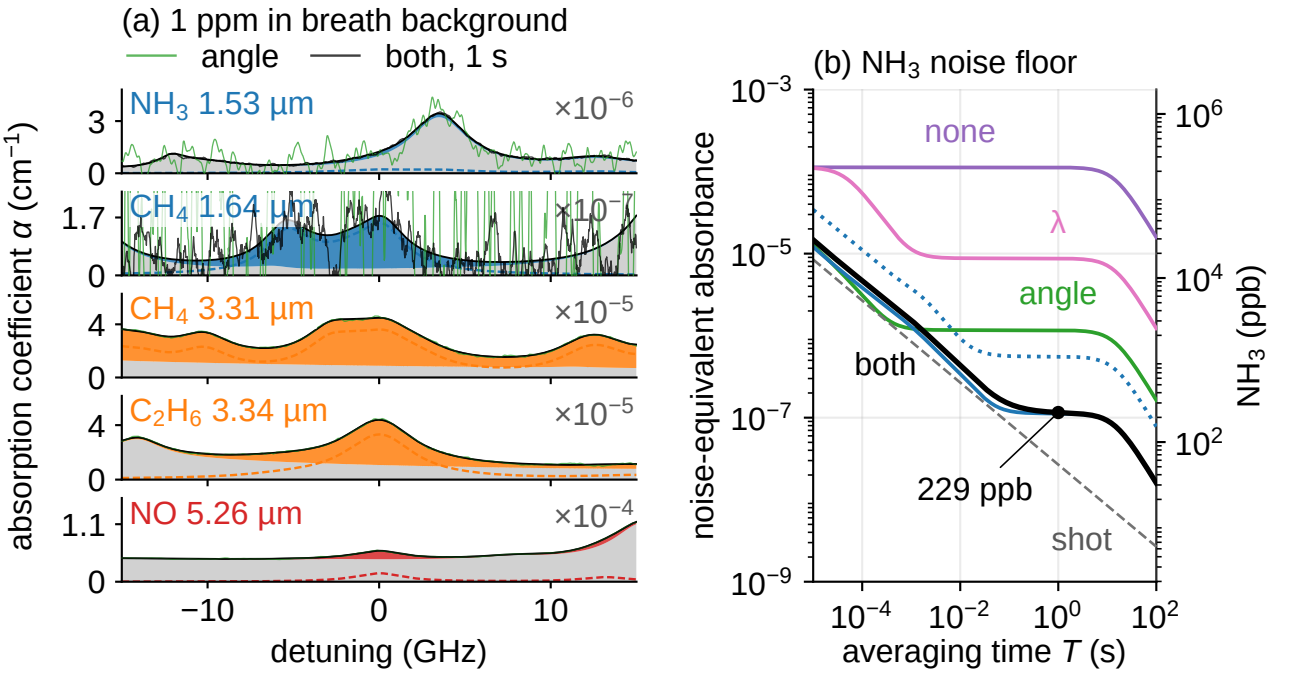

(a) 1 ppm in breath background
angle
both, 1 s
NH3 1.53 μm
×10−6
3
0
CH4 1.64 μm
×10−7
1.7
0
CH4 3.31 μm
×10−5
4
0
C2H6 3.34 μm
×10−5
4
0
NO 5.26 μm
×10−4
1.1
0
absorption coefficient α (cm−1)
−10
0
10
detuning (GHz)
(b) NH3 noise floor
noise-equivalent absorbance
10−3
10−5
10−7
10−9
none
λ
angle
both
229 ppb
shot
10−4
10−2
100
102
averaging time T (s)
106
104
102
NH3 (ppb)